# Design and Analysis of 4-bit Flash ADC using CMOS 180nm Technology.

*DAU, Gandhinagar, India*

Deepak Chidambaran, Rutu Parekh

202411028@daiict.ac.in, rutu_parekh@daiict.ac.in

***Abstract*— The performance of Flash Analog-to-Digital converter is greatly influenced by the choice of Comparator and Thermometer-to- Binary encoder design. The work describes the design and pre-simulation of a 4bit analog to digital converter for low power CMOS. It requires 2N -1 comparators, an encoder to convert thermometer code to binary code. The design is simulated in cadence environment using spectre simulator under 180nm technology. Each comparator in the array evaluates the input signal by comparing it with reference voltages derived from a resistor ladder. The implemented design demonstrates a power consumption of 49.77 nW while operating at a supply voltage of 1.8V. The circuit achieves an operational frequency of 100 MHz.**

***Keywords— Analog-to-Digital converter, Thermometer-to-binary encoder, comparator, resistive- ladder.***

## I. Introduction

Analog-to-Digital Converters (ADCs) serve a fundamental role in transforming continuous-time, continuous-amplitude analog signals into digital signals characterized by discrete time intervals and quantized amplitude levels. These digital levels are typically represented as powers of two (e.g., 2, 4, 8, 16), making the binary system—with only two possible states—an ideal match for digital electronics. An ADC receives analog voltages that can assume a continuous range of values and produces a digital output composed of distinct numerical states. Among the different ADC architectures, the flash ADC stands out for its extremely high sampling rates. Its speed is predominantly determined by the performance of its comparators, which in turn are influenced by the regeneration time constant—a parameter inversely related to the square of the transistor gate length in CMOS processes. Several ADC types are optimized for various applications. Flash ADCs offer the fastest operation, Successive Approximation Register (SAR) ADCs strike a balance between speed and resolution, Delta-Sigma ADCs are used where high resolution is necessary, and Dual-Slope ADCs are preferred for their precision, albeit with slower conversion times. In the proposed 4-bit flash ADC, an optimized architecture uses fewer transistors, leading to improved speed, lower power usage, and reduced silicon footprint when compared to traditional designs. A layout diagram for the block structure is included in Section IV. These ADCs are particularly suitable for high-speed digital communication systems—such as radar, video processing, and data acquisition—where rapid signal digitization is more critical than high resolution.

The architecture includes three primary blocks: a sample-and-hold circuit, a comparator array, and a digital encoder. The sample-and-hold unit stabilizes the input signal, which is then processed by the comparators to generate a thermometer-coded output. A New Modified Pre-amplifier latch comparator architecture is employed, allowing for a reduced comparator count—approximately half of what traditional flash ADCs require. Specifications for this 4-bit flash ADC, designed using 180 nm CMOS technology, include operation at 1 V, a maximum input swing of 1.8 V peak-to-peak, and support for sampling rates up to 100 MHz, as summarized in Table I.

TABLE I. Specification in the input

| | |
|---|---|
| Technology | 180nm CMOS |
| Operating Voltage | 1 V |
| Input Voltage | 1.8 V |
| Resolution | 4 - bits |
| Sampling Frequency | 100MHz |

## II. Flash ADC ARCHITECTURE

Flash ADCs, often referred to as parallel ADCs, are among the fastest types of converters used to translate analog signals into digital format. Despite their impressive speed, they come with certain limitations, including lower resolution and greater circuit complexity, which in turn lead to increased silicon area usage. However, they still maintain advantages over other ADC architectures in terms of speed and power efficiency. Fig. 1 presents the full schematic of the proposed 4-bit flash ADC. Below is an explanation of the structure and functionality of this architecture:

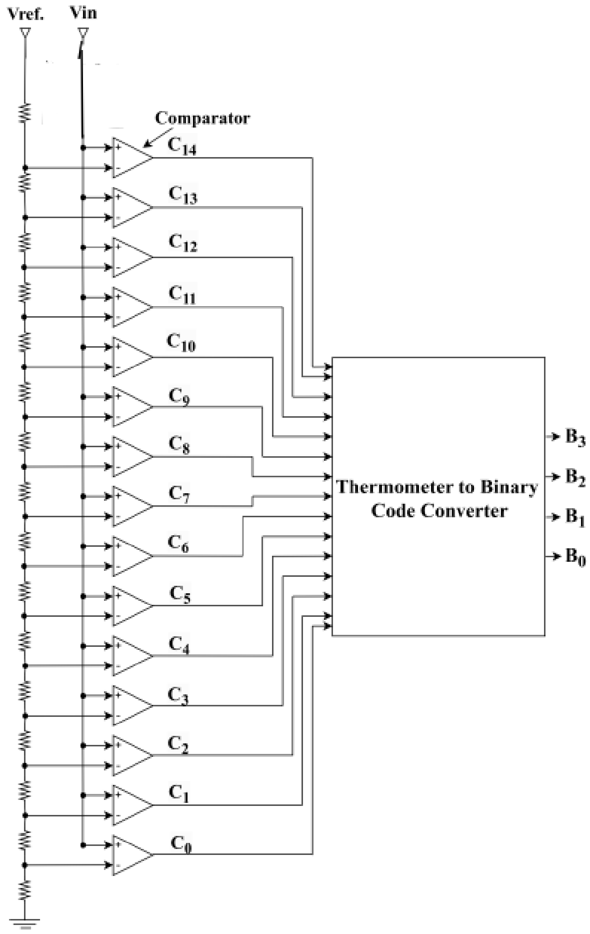


Fig.1. 4-bit flash ADC Architecture.

1. Resistive Ladder: This section generates multiple reference voltages, which serve as benchmarks to compare against the incoming analog signal. These voltages help determine the digital equivalent of the input.
2. Comparator : At the core of a flash ADC lies an array of comparators. Each comparator evaluates the analog input against a specific reference voltage from the ladder. The number of comparators required is 2^n−1, where *n* is the desired output resolution. For instance, a 4-bit ADC includes 15 comparators.
3. Reference Voltage Circuit: A precision voltage reference unit provides the necessary voltage levels for each comparator. These values generally follow a binary-weighted or evenly spaced pattern to ensure consistent signal comparison.
4. Thermometer-to-Binary Conversion: Comparator outputs, arranged in a thermometer code (a string of consecutive '1's followed by '0's), are processed by a code converter. This block translates the thermometer-coded signal into a more compact binary output. The conversion logic is demonstrated in Fig3.
5. Digital Output Stage: The binary output from the code converter represents the final digital form of the analog input signal. Each bit of this output corresponds to a specific weight, reflecting the magnitude of the analog input.

## III. Methodology

To design a 4 -bit flash ADC, the architecture is divided into three main section: The Comparator block, Resistive adder and Encoder Block. Each section plays an important and vital role in converting analog signals into digital outputs.

### A. *Comparator*

The Flash ADC design incorporates a low-power comparator circuit, as illustrated in Fig. 2 and originally introduced in [1]. This comparator architecture features two primary stages: a preamplifier and a latch. In the preamplifier section, NMOS transistors form the differential input pair, while PMOS devices serve as the load, enabling sufficient voltage gain. The latch stage comprises a pair of cross-coupled inverters, forming a differential comparator configuration. Additionally, an NMOS transistor is placed between the latch's differential nodes to enhance switching performance. The comparator design was developed and simulated using Cadence tools in a 180nm CMOS process [1].

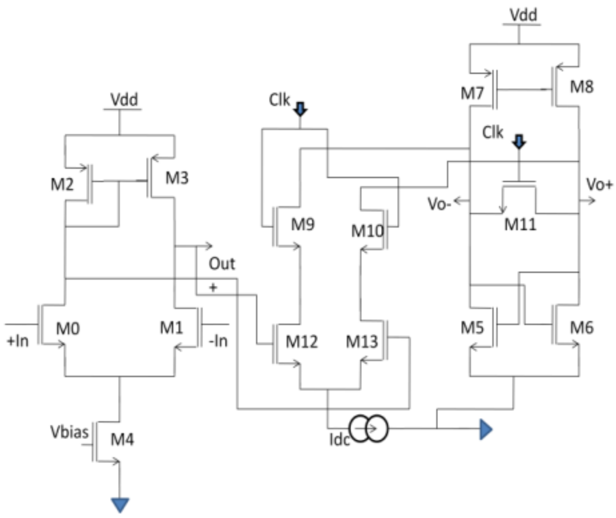


Fig.2. Circuit of Preamplifier-Latch comparator

In the present work, pre-simulation results are also obtained under the same 180nm technology. The preamplifier not only increases the comparator's sensitivity but also shields the input from disturbances caused by the regenerative feedback stage [3]. The latch, on the other hand, determines which input signal is higher and boosts the voltage difference between them for accurate digital interpretation [7].

### B. *Resistive Ladder*

In an Analog-to-Digital Converter (ADC), the resistor ladder plays a crucial role in supplying a series of steady reference voltages to the array of comparators. For a digital input of *n* bits (such as a 4-bit input), each bit corresponds to a switch that connects either to a reference voltage (Vref) or to ground. The resistor ladder, generally made up of 2^N resistors, creates a sequence of reference voltages that are evenly spaced. These voltages form a binary-weighted sequence, with each comparator receiving a reference value that is one Least Significant Bit (LSB) lower than the one provided to the comparator directly above it in the sequence. This configuration allows every comparator to receive a distinct reference level, ensuring precise comparison of the incoming analog signal and enabling accurate digital conversion.

### C. *Thermometer to Binary code converter*

The logic encoder used for 4-bit ADC is shown in Fig. 3. It is a multiplexer based encoder which converts thermometer codes to binary codes. Table 2 shows the truth table for 4-bit encoders. The multiplexers are designed by using transmission gates.

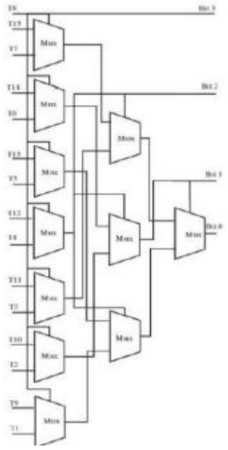

Fig.3. 4-bit logic encoder

TABLE II. Truth Table for 4-bit encoder

| T15 | T14 | T13 | T12 | T11 | T10 | T9 | T8 | T7 | T6 | T5 | T4 | T3 | T2 | T1 | B4 | B3 | B2 | B1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

The Boolean expression can be observed here as:

$G_3 = T_8$

$G_2 = T_4\overline{T}_{12}$

$G_1 = T_2\overline{T}_6 + (T_{10}\overline{T}_{14})T_6$

$G_0 = T_1\overline{T}_3 + (T_5\overline{T}_7 + (T_9\overline{T}_{11} + (T_{13}\overline{T}_{15})T_{11})T_7)T_3$

The code converting from gray to binary requires an XOR

B4 = G3

B3 = XOR of (B2,G2)

B2 = XOR of (B2,G1)

B1 = XOR of (B1,G0)

## IV. Simulation Results

### A. *Comparator*

The schematic of a Pre-amplifier – Latch comparator using 180nm technology is shown in Fig. 4. The input signal applied to the non-inverting terminal is a sine wave and a dc reference voltage is applied to the inverting terminal of the comparator. The output waveform of the given comparator as shown in Fig .5.

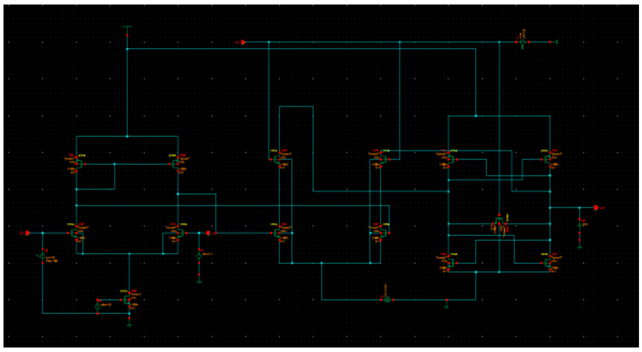

Fig .4. Schematic of Preamplifier – Latch comparator

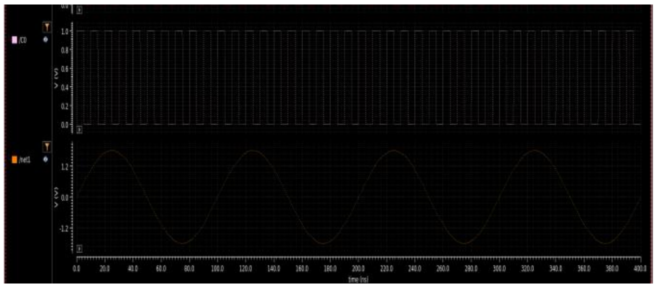

Fig .5. Output Waveform of the Comparator

### B. *Flash ADC*

The above obtained 2^N -1 comparator outputs are encoded into output for 4- bit ADC respectively using an encoder and the resulting waveform of resolution ADC are as shown in Fig.6 and Fig.7.

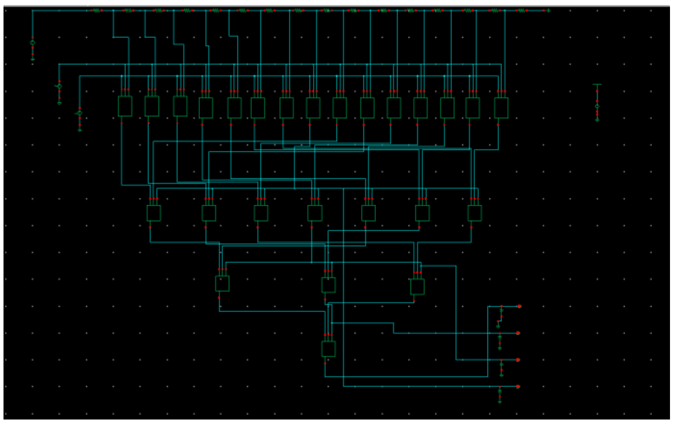

Fig.6. Schematic of 4 -bit Flash ADC.

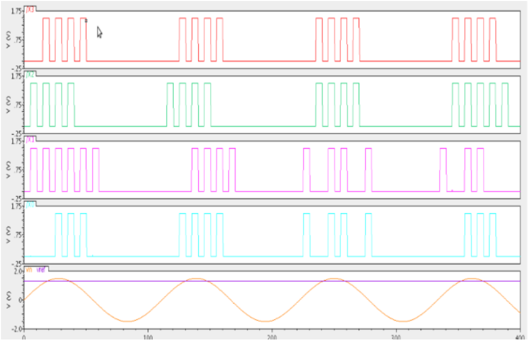

Fig. 7. Output Waveform of 4 -bit FLASH ADC.

TABLE III. RESULTS

| Parameter | Value |
|---|---|
| Delay | 24.66ns |
| Average Power | 49.77nW |
| ENOB (bits) | 3.57 |
| SNDR (dB) | 22.65 |

Table III presents a concise overview of the system's core performance parameters, including power consumption, speed, and signal integrity. The design operates with an average power usage of 49.77 nW, system exhibits a delay of 24.66 ns, indicating its temporal response. In terms of resolution, the Effective Number of Bits (ENOB) is calculated to be 3.57. Signal fidelity is characterized by a Signal-to-Noise and Distortion Ratio (SNDR) of 22.65 dB, which measures output clarity. These measures offer a comprehensive insight into the system's overall performance.

## V. Conclusion

In this flash ADC design, the use of fewer transistors contributes to enhanced operating speed and reduced average power consumption compared to conventional ADC architectures. With further tuning and development, the performance can be improved even more, enabling its integration into a wider range of electronic systems. The architecture demonstrates rapid conversion capabilities, making it well-suited for high-speed digital applications. Utilizing a resistor ladder for reference voltage generation and use of 2:1 Multiplexer in the encoder circuit, the design effectively converts thermometer-coded signals into binary output with improved efficiency.

## VI. Referrences


[1] L. Koutanali, S. S. Kotabagi, R. Vernekar, V. D. Chitragar, and P. Sangalad, "3-bit Flash ADC using TIQ Comparator," in *Proc. 2024 5th Int. Conf. Emerging Technol. (INCET)*, Belgaum, India, 2024, pp. 1–6, doi: 10.1109/INCET61516.2024.10593264.

[2] H.-H. Thai, C.-K. Pham, and D.-H. Le, "Design of a High-Speed 8-bit Flash ADC using Double-Tail Comparator on 180nm CMOS Process," in *Proc. 2021 8th NAFOSTED Conf. Inf. Comput. Sci. (NICS)*, Hanoi, Vietnam, 2021, pp. 329–334, doi: 10.1109/NICS54270.2021.9701505.

[3] L. R. Thoutam et al., "Design of Low Power 2-Bit Flash ADC using High Performance Dynamic Double Tail Comparator," in *Proc. 2022 IEEE Int. Conf. Nanoelectron., Nanophoton., Nanomater., Nanobiosci. Nanotechnol. (5NANO)*, Kottayam, India, 2022, pp. 1–5, doi: 10.1109/5NANO53044.2022.9828927.

[4] P. Sowmya, M. Samson, and M. J. Mehdi, "Design of Two Stage Operational Amplifier and Implementation of Flash ADC," in *Proc. 2021 3rd Int. Conf. Intell. Commun. Technol. Virtual Mobile Netw. (ICICV)*, Tirunelveli, India, 2021, pp. 490–496, doi: 10.1109/ICICV50876.2021.9388589.

[5] T. C. Dos Santos, A. G. Girardi, and P. C. C. De Aguirre, "A Low-Power 4-Bit 20-MS/s Flash ADC in 65-nm CMOS with Rail-to-Rail Comparators," in *Proc. 2024 Argentine Conf. Electron. (CAE)*, Bahía Blanca, Argentina, 2024, pp. 103–108, doi: 10.1109/CAE59785.2024.10487165.

[6] S. Shahramian, S. P. Voinigescu, and A. C. Carusone, "A 35-GS/s, 4-Bit Flash ADC With Active Data and Clock Distribution Trees," *IEEE J. Solid-State Circuits*, vol. 44, no. 6, pp. 1709–1720, Jun. 2009, doi: 10.1109/JSSC.2009.2020657.

[7] G. M. et al., "High Speed and Low Power Modified Dynamic Double Tail Comparator on 90nm Technology using Cadence," in *Proc. 2022 Int. Conf. Power, Energy, Control Transm. Syst. (ICPECTS)*, Chennai, India, 2022, pp. 1–7, doi: 10.1109/ICPECTS56089.2022.10047598.

[8] C. Huang, C. Wang, and J. Wu, "A CMOS 6-Bit 16-GS/s Time-Interleaved ADC Using Digital Background Calibration Techniques," *IEEE J. Solid-State Circuits*, vol. 46, no. 4, pp. 848–858, Apr. 2011.

[9] C. H. Chan et al., "A 5-Bit 1.25-GS/s 4×-Capacitive-Folding Flash ADC in 65-nm CMOS," *IEEE J. Solid-State Circuits*, vol. 48, no. 9, pp. 2154–2169, Sep. 2013.

[10] J.-I. Kim, B.-R.-S. Sung, W. Kim, and S.-T. Ryu, "A 6-b 4.1-GS/s Flash ADC With Time-Domain Latch Interpolation in 90-nm CMOS," *IEEE J. Solid-State Circuits*, vol. 48, no. 6, pp. 1429–1441, Jun. 2013, doi: 10.1109/JSSC.2013.2252516.

[11] M. R. Megha and K. A. Pradeepkumar, "Implementation of Low Power Flash ADC by Reducing Comparators," in *Proc. 2014 Int. Conf. Commun. Signal Process. (ICCSP)*, Melmaruvathur, India, 2014, pp. 443–447, doi: 10.1109/ICCSP.2014.6949880.

[12] S. Hussain, R. Kumar, and G. Trivedi, "Comparison and Design of Dynamic Comparator in 180nm SCL Technology for Low Power and High Speed Flash ADC," in *Proc. 2017 IEEE Int. Symp. Nanoelectron. Inf. Syst. (iNIS)*, Bhopal, India, 2017, pp. 139–144, doi: 10.1109/iNIS.2017.37.

[13] A. Payra, P. Dutta, A. Sarkar, and S. K. Sen, "Design of a Self Regulated Flash Type ADC with High Resolution," in *Proc. Michael Faraday IET Int. Summit*, Kolkata, India, 2015, pp. 591–595, doi: 10.1049/cp.2015.1698.

[14] P. V. Lakshmi, S. Musala, and A. Srinivasulu, "Flash ADC's Low Power, High Speed Dynamic Comparator," in *Proc. 2022*

*IEEE Int. Women Eng. (WIE) Conf. Electr. Comput. Eng. (WIECON-ECE)*, Naya Raipur, India, 2022, pp. 352–355, doi: 10.1109/WIECON-ECE57977.2022.10151081.

[15] A. Khorami, M. B. Dastjerdi, and A. F. Ahmadi, “A Low-Power High-Speed Comparator for Analog to Digital Converters,” in *Proc. 2016 IEEE Int. Symp. Circuits Syst. (ISCAS)*, Montreal, QC, Canada, 2016, pp. 2010–2013, doi: 10.1109/ISCAS.2016.7538971.

[16] S. Nanda and A. S. Panda, “Design of Conventional Three-Stage CMOS Comparator in 90-nm CMOS Technology and Comparative Analysis with Its Counterparts,” in *Proc. 2015 Int. Conf. Smart Sensors Syst. (IC-SSS)*, Bangalore, India, 2015, pp. 1–6, doi: 10.1109/SMARTSENS.2015.7873613.

[17] K. Thakur and S. K. Kingra, “Design and Implementation of Hybrid 4-bit Flash ADC,” in *Proc. 2016 Int. Conf. Adv. Comput., Commun. Informat. (ICACCI)*, Jaipur, India, 2016, pp. 1233–1237, doi: 10.1109/ICACCI.2016.7732214.

[18] M. R. Megha and K. A. Pradeepkumar, “Implementation of Low Power Flash ADC by Reducing Comparators,” in *Proc. 2014 Int. Conf. Commun. Signal Process. (ICCSP)*, Melmaruvathur, India, 2014, pp. 443–447, doi: 10.1109/ICCSP.2014.6949880.